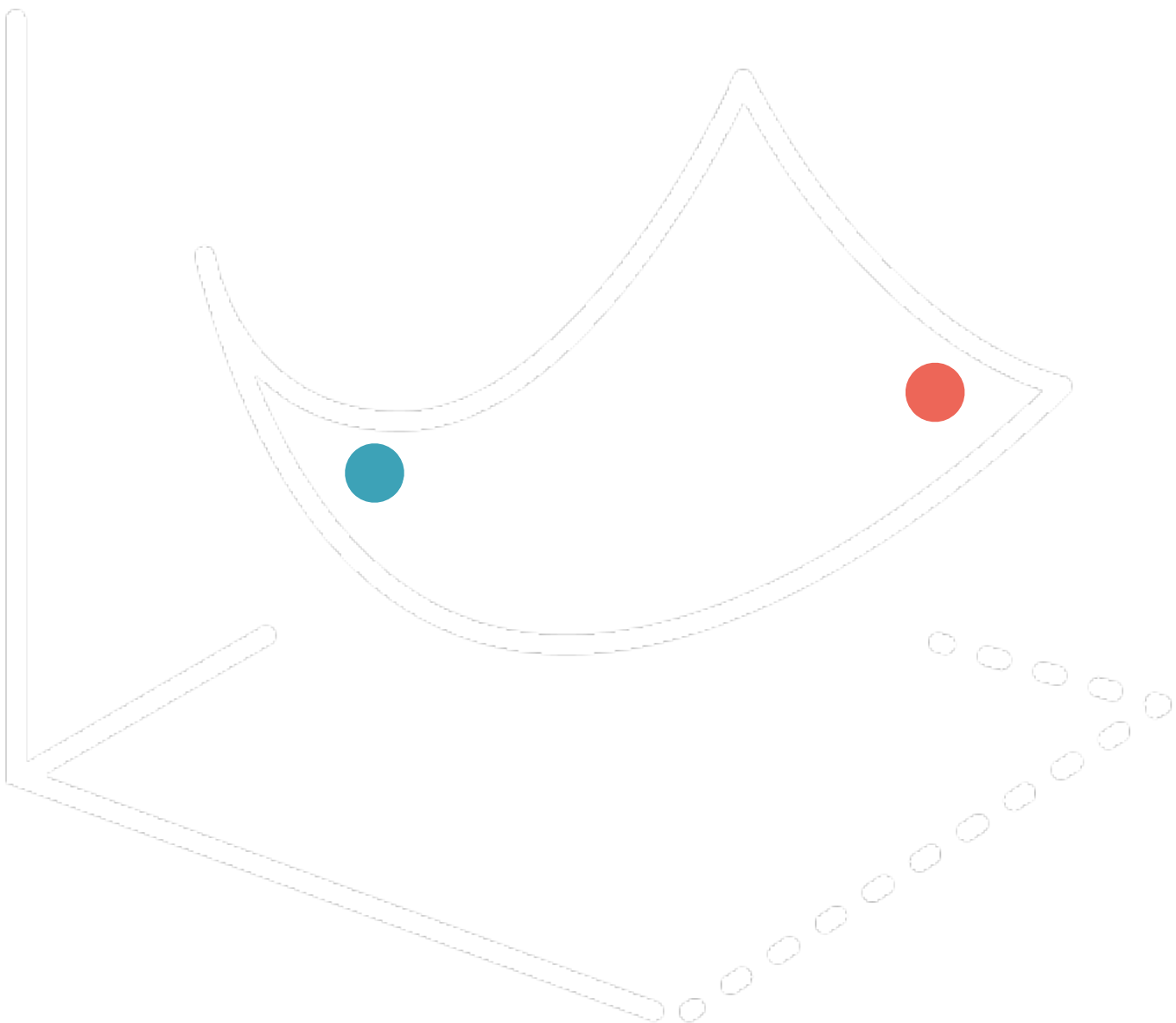


Probability measures belief in a metric space.

Provides natural stage for ML tasks (datasets $\leftrightarrow$ measures).

α

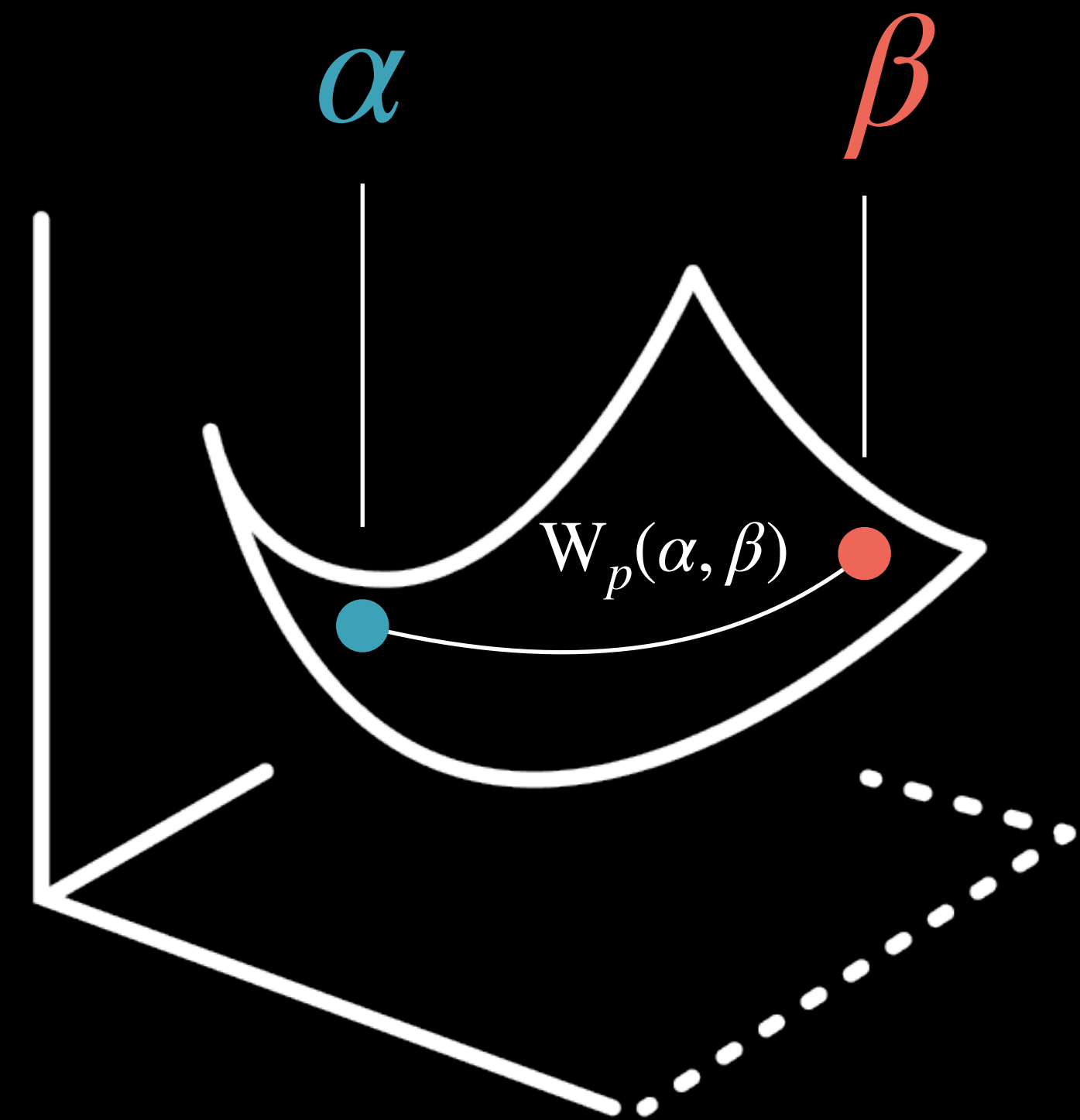
β



The Wasserstein Space

The Wasserstein Space $\mathbb{W}_p$

- We can endow $\mathcal{P}(\mathcal{X})$ with a the metric W_p .
- This yields $\mathbb{W}_p := (\mathcal{P}(\mathcal{X}), W_p)$, the p-Wasserstein space (a metric space).
- $\mathbb{W}_p$ has useful properties: complete, separable, existence of geodesics, etc
- Vast theory characterizing various properties of $\mathbb{W}_p$ (eg Villani, etc)
- Why is this important?
 - **Probability measures become points in a metric space.**
 - Opens up geometry & analysis: gradient flows, barycenters.
 - **Provides a natural stage for ML tasks (datasets $\leftrightarrow$ measures).**



The Wasserstein Space

OT as a lifting operation

- Given a metric space $(\mathcal{X}, d)$ where data lives, and using OT with cost $= d$
- OT can be thought of as *lifting* the point-wise metric d into a metric between distributions on that space.